

Yuhan Wang

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Education

ETH Zürich <i>Doctor of Science in Geophysics, expected Dec. 2026; doctoral coursework average: 5.7/6.0</i> (Supervisors: Prof. Elías R. Heimisson and Prof. Stefan Wiemer)	Jan. 2023 – Dec. 2026 Zürich, Switzerland
Tongji University <i>M.Eng. in Geological Engineering, GPA: 92.0/100</i> (Advisor: Prof. Wuwei Mao)	Sep. 2019 – Mar. 2022 Shanghai, China
Central South University <i>B.Eng. in Civil Engineering; GPA: 88/100</i>	Sep. 2015 – Jun. 2019 Changsha, Hunan, China

Experience

Stanford University <i>Visiting Student Researcher in Geophysics</i>	Jul. 2025 – Jun. 2026 California, USA
University of Iceland <i>Visiting Student in Earth Sciences</i>	Aug. 2024 – Sep. 2024, Mar. 2025 – May 2025 Reykjavik, Iceland
ETH Zürich <i>Scientific Assistant</i>	Jan. 2023 – Present Zürich, Switzerland
University of Warwick <i>Research Assistant in Computational Geomechanics</i>	Jul. 2022 – Oct. 2022 Remote
Tongji University <i>Research/Teaching Assistant</i>	Sep. 2019 – Jul. 2022 Shanghai, China
Monash University <i>Visiting Research Student</i>	Dec. 2018 – May 2019 Melbourne, Victoria, Australia

Publications

(* Corresponding author)

1. **Yuhan Wang***, Elías Rafn Heimisson. Poroelastic bulk controls on shear localization in fault gouge during earthquake rupture: Insights from a hybrid numerical framework. *Journal of Geophysical Research: Solid Earth*, 131(3), e2025JB032291 (2026). doi: 10.1029/2025JB032291.
2. **Yuhan Wang**, Guotao Ma, Mohammad Rezaia. A granular anisotropic model of underground rockburst considering the effect of radial stresses. *Tunnelling and Underground Space Technology*, 155, 106202 (2025). doi: 10.1016/j.tust.2024.106202.
3. **Yuhan Wang***, Elías Rafn Heimisson. A coupled finite difference–spectral boundary integral method with applications to fluid diffusion in fault structures. *International Journal for Numerical and Analytical Methods in Geomechanics*, 48(10), 2564–2591 (2024). doi: 10.1002/nag.3740.
4. Wuwei Mao, **Yuhan Wang***, Ping Yang, Yu Huang, Hu Zheng. Dynamics of granular debris flows against slit dams based on the CFD–DEM method: effect of grain size distribution and ambient environments. *Acta Geotechnica*, 18, 5811–5838 (2023). doi: 10.1007/s11440-023-01944-y.
5. **Yuhan Wang**, Nhu H. T. Nguyen. The effects of rock–infill interfacial properties on the compressive damage behaviour of flawed rocks: a DEM study. *Theoretical and Applied Fracture Mechanics*, 117, 103166 (2022). doi: 10.1016/j.tafmec.2021.103166.
6. **Yuhan Wang**, Nhu H. T. Nguyen, Lianheng Zhao. Micromechanical study on hard rock strainburst using the discrete element method. *Tunnelling and Underground Space Technology*, 109, 103793 (2021). doi: 10.1016/j.tust.2020.103793.
7. Ping Yang, Zixin Zhao, Zhicheng Li, **Yuhan Wang**. Experimental study on long-term performance of new urban green space soil for sponge city construction. *Urban Forestry & Urban Greening*, 58, 126906 (2021). doi: 10.1016/j.ufug.2020.126906.

Publications (continued)

8. **Yuhan Wang**, Ping Yang, Zhitang Li, Shaojie Wu, Zixin Zhao. Experimental–numerical investigation on grout diffusion and washout in rough rock fractures under flowing water. *Computers and Geotechnics*, 126, 103717 (2020). doi: 10.1016/j.compgeo.2020.103717.

Manuscripts Under Review and in Preparation

1. Xiaokun Xie, **Yuhan Wang**, Pan Li, Shaoshuai Shi, Xiaoli Rong, Weitao Wu, Jiangzhou Peng, Xintong Wang, Jie Hu. Dynamic generalized Hoek-Brown constitutive model for rock mass. Manuscript under review at *Rock Mechanics and Rock Engineering*.
2. **Yuhan Wang***, Elías Rafn Heimisson, Eric Dunham. Understanding localization and delocalization in earthquake cycles using a multiscale rate-and-state friction framework. Manuscript in preparation.

Oral Presentations

1. **Yuhan Wang**, Elías Rafn Heimisson. Poroelastic Controls on Shear Localization in Fault Gouge: A Cross-Scale Hybrid Numerical Framework. *2026 Crustal Deformation Modeling Workshop, Colorado School of Mines, Golden, Colorado, USA*.
2. **Yuhan Wang**, Elías Rafn Heimisson. Poroelastic Bulk Controls on Shear Localization in Fluid-Saturated Fault Gouge During Earthquake Rupture. *AGU Fall Meeting 2025, New Orleans, USA*.
3. **Yuhan Wang**, Elías Rafn Heimisson. Poroelastic Control of Shear Localization in Fluid-Saturated Fault Zones. *SCEC Meeting 2025, Palm Springs, USA*.
4. **Yuhan Wang**, Elías Rafn Heimisson. Investigating the Influence of Poroelastic Bulk on Small-scale Shear Localization with a Hybrid SBI-FD Method. *AGU Fall Meeting 2024, Washington, D.C., USA*.
5. **Yuhan Wang**. Understanding Shear Localization within Fault Gouge with Hybrid Numerical Schemes. *Weekly Seminar of Prof. David Kammer's Group, Nov. 2024, Institute for Building Materials, ETH Zürich, Zürich, Switzerland*.
6. **Yuhan Wang**. Unveiling Fault-Fluid Interactions with Hybrid Numerical Schemes. *Friday Seminar of the Institute of Earth Sciences and Nordic Volcanological Center, 13 Sep. 2024, University of Iceland, Reykjavik, Iceland*.
7. **Yuhan Wang**, Elías Rafn Heimisson. Modeling of Shear Localization of Fault Gouge with a Hybrid Spectral Boundary Element and Finite Difference Approach. *EGU General Assembly 2024, Vienna, Austria*.
8. **Yuhan Wang**, Wuwei Mao, Elías Rafn Heimisson. On the effect of vibration on dry/fluid-saturated granular flows: Implications for geological hazards induced by earthquakes. *EGU General Assembly 2023, Vienna, Austria*.
9. **Yuhan Wang**, Wuwei Mao, Ping Yang, Yu Huang, Hu Zheng. CFD-DEM study on the entrainment induced by debris flows with the HBP rheological model. *11th Conference of Asian Rock Mechanics Society 2021, Beijing, China*.

Awards & Distinctions

SCEC Travel Grant , Statewide California Earthquake Center	2025
Doctoral Mobility Fellowship , Swiss National Science Foundation	2024
Best Master's Thesis Award , Tongji University	2022
Outstanding Graduate Student Award , Tongji University	2022
National Scholarship , Tongji University	2020, 2021
Graduate Full Scholarship , Tongji University	2019–2022
International Exchange Scholarship for Undergraduates , Central South University	2018
First Prize , China Undergraduate Physics Tournament (Central South University)	2018
Jiang Weiying Scholarship , awarded by a Hong Kong entrepreneur	2017